

**NEW YORK CITY COLLEGE OF TECHNOLOGY
COMPUTER SYSTEMS TECHNOLOGY DEPARTMENT
CST4714 - DATABASE ADMINISTRATION
(2 class hours, 2 lab hours, 3 credits)**

Instructor: Atilio Barreda II **Email:** abarreda@citytech.cuny.edu

COURSE DESCRIPTION: Students in this course will develop a fundamental understanding of the tasks and issues associated with database administration, including planning, building, tuning, troubleshooting, securing, and monitoring databases. Students will learn how to manage users, privileges, and resources, implement basic backup and recovery procedures and identify tuning opportunities. Students will practice these responsibilities using managed cloud platforms (Supabase for PostgreSQL and MongoDB Atlas for MongoDB) while learning the underlying concepts needed to diagnose performance, reliability, and security issues. All major assignments and projects can be completed using free-tier, browser-based tools.

COURSE OBJECTIVES: Upon successful completion of this course, the students will have acquired the following knowledge and skills:

1. Understanding the problems and appropriate solutions involved in the role of database management and administration
2. Installing, creating, configuring, and maintaining an operational database
3. Managing physical and logical database structures, including users, objects, and data
4. Monitoring performance and identifying performance improvements
5. Planning and implementing backup/recovery strategies
6. Managing database security

REQUIRED TEXTBOOK

No paid textbook is required. Readings will be assigned from official documentation and instructor materials posted on GitHub.

Required resources (free):

- PostgreSQL Official Documentation (SQL, Administration, Performance/EXPLAIN)
- Supabase Documentation (SQL Editor, Auth, Row Level Security, migrations)
- MongoDB Manual and MongoDB Atlas Documentation
- Instructor GitHub materials (weekly notes, lab handouts, starter repos)

ADDITIONAL MATERIAL

The instructor will identify several additional information resources during the semester including Internet resources and print material (handouts). The instructor will also publish educational material (guides, sample code, lab templates) on GitHub throughout the semester.

ACADEMIC INTEGRITY POLICY:

Students and all others who work with information, ideas, texts, images, music, inventions, and other intellectual property owe their audience and sources accuracy and honesty in using, crediting, and citing sources. As a community of intellectual and professional workers, the College recognizes its responsibility for providing instruction in information literacy and academic integrity, offering models of good practice, and responding vigilantly and appropriately to infractions of academic integrity. Accordingly, academic dishonesty is prohibited in The City University of New York and at New York City College of Technology and is punishable by penalties, including failing grades, suspension, and expulsion. The complete text of the College policy on Academic Integrity may be found in the catalog.

GRADING PROCEDURE:

Midterm Project	30%
Final Project	40%
Engagement/Participation	30%
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TOTAL	100%

Letter Grade	A	A-	B+	B	B-	C+	C	D	F
Numerical Grade	93-100	90-92.9	87-89.9	83-86.9	80-82.9	77-79.9	70-76.9	60-69.9	<=59.9

COURSE OUTLINE:

Week	Topics	Reading
1	Database administration - tasks and roles Tooling overview (Supabase, Atlas, GitHub/Codespaces)	Instructor GitHub notes; Supabase quickstart; Atlas quickstart
2	PostgreSQL fundamentals from scratch Basic SQL: SELECT/WHERE/JOIN/GROUP BY	PostgreSQL docs: SQL language; data definition
3	Database structures (Postgres): schemas, tables, indexes In-class reflection	PostgreSQL docs: indexes; instructor handout
4	Schema Management (Postgres) Views, sequences, constraints; introspection	PostgreSQL docs: DDL; views; sequences
5	Managing Data and Concurrency (Postgres) Transactions (ACID), isolation, locking symptoms	PostgreSQL docs: transactions; locking
6	User management and security (Postgres + Supabase) - Roles/grants + Supabase Auth + RLS - In-class reflection (credit/no credit)	Supabase docs: RLS; PostgreSQL docs: roles
7	Performance (Postgres): EXPLAIN + query plans (intro) Indexing strategy; slow query triage	PostgreSQL docs: EXPLAIN; performance
8	Backup and Recovery (Postgres/Supabase) - Restore-first mindset; safe migrations - Midterm Project work session - Midterm Presentations (part 1)	PostgreSQL docs: backup/restore; Supabase backups
9	Midterm Presentations (part 2) MongoDB introduction (documents, collections, CRUD)	MongoDB Manual: CRUD; Atlas docs
10	MongoDB data modeling (embed vs reference) - Schema validation; indexing basics - In-class reflection (credit/no credit)	MongoDB Manual: data modeling; indexes
11	Aggregation pipeline + performance troubleshooting Using explain/profiler; common anti-patterns	MongoDB Manual: aggregation; performance
12	Replication and reliability concepts (Atlas) Read preference + write concern; backup/restore	MongoDB Manual: replication; Atlas backups
13	Scaling concepts: sharding + capacity planning - Shard keys, hotspots, cost signals - In-class reflection (credit/no credit)	MongoDB Manual: sharding

14	Advanced topics: polyglot persistence + distributed tradeoffs • Final Project work session + peer review	Instructor GitHub handout
15	Final Project presentations • Course wrap-up: portfolio packaging + interview review	None

CLOUD AND COMPUTING RESOURCES

This course is designed so that students do not need a personal laptop. All required work can be completed using free, browser-based tools and campus computers (if desired). Primary platforms used in this course:

- Supabase (free tier) - managed PostgreSQL with a browser SQL editor and authentication/RLS features
- MongoDB Atlas (free tier M0) - managed MongoDB with a browser UI, aggregation builder, and backups
- GitHub (free) - submissions and version control; materials will be published on GitHub

IN-CLASS REFLECTIONS

Some class meetings will include short in-class writing assignments responding to a brief reading, video, or case study (for example: a postmortem, a data breach write-up, or a short technical article). These reflections are not formally graded; students receive participation credit for completing them in class. In case of absences, reflections can be made up.

PROJECTS

All projects in this course are individual. Each project includes a short in-class presentation.

Midterm Project (PostgreSQL + Supabase) - 30%

- Design and implement a relational schema for a small application domain
- Write SQL migrations and seed data; demonstrate basic analytics queries
- Implement role-based access controls and (when applicable) Supabase Row Level Security policies
- Perform a simple performance investigation (EXPLAIN + indexing change) and document results
- Write a restore-first runbook: backup/restore steps and a quick verification checklist
- Deliverables: GitHub repo + brief written report; 5-minute presentation (Weeks 8-9)

Final Project (MongoDB + Atlas) - 40%

- Design and implement a MongoDB data model (embed vs reference) for a workload
- Create indexes and demonstrate at least one aggregation pipeline used for analysis/reporting
- Use explain/profiler (or Atlas performance tools) to identify and resolve a performance issue
- Describe reliability choices (write concern/read preference) and perform a backup/restore verification
- Deliverables: GitHub repo + brief written report; 7-minute presentation (Week 15)

ASSESSMENT CRITERIA:

For the successful completion of this course, a student should be able to demonstrate understanding of key database concepts and the ability to apply them in practical administration tasks, including but not limited to:

1. Database management and administration roles
2. Database installation, creation, configuration, and maintenance issues (local optional; cloud required)
3. Management of physical and logical data structures, including users, objects, and data
4. Monitoring performance and improving performance
5. Backup/recovery planning and restore drills
6. Database security issues (least privilege; auditing; row-level controls in PostgreSQL/Supabase)

Evaluation methods: quizzes, homework assignments, in-class lab exercises, and case study/projects.